

CS471 – Web Technologies (Laboratory)		Lab 10
		Django Forms (Part 1)

This lab session covers Django Forms (Part 1 – Create, read (display or list), update, and delete). By the end of this session, students should have a very good understanding of how to build CRUD operations in Django.

Pre-lab Preparation:

1. The table book as in lab7.
2. The fake data from mockaroo.com website or any other methods.

Lab Activities:

Part 1: CRUD operations and Template organization (without using Django forms)

Task 1: Create a URL `'/books/lab9_part1/listbooks'` with any necessary HTML file and view function to list books and provide a link to add a new book (implemented in the following task (Task 2)). Also, two buttons/links are associated with each book that hold the book's ID to edit or delete the book (implemented in Task 3 and Task 4).

```

> static
  templates
    bookmodule
      aboutus.html
      booklist.html
      formatting.html
      index.html
      lab9_task1.html
      172 def list_books_lab10(request):
      173     books = Book.objects.all()
      174     return render(request, 'bookmodule/lab10_list.html', {'books': books})
      175 def add_book_lab10(request):
      176     return render(request, 'bookmodule/lab10_addbook.html')
      177 def edit_book_lab10(request, id):
      178     return render(request, 'bookmodule/lab10_editbook.html')
      179 def delete_book_lab10(request, id):
      180     return render(request, 'bookmodule/lab10_deletebook.html')

lab9_task1.html 30
lab9_task2.html 31
lab9_task3.html 32
lab9_task4.html 33
lab10_part1/listbooks, views.list_books_lab10,
lab10_part1/addbook, views.add_book_lab10,
lab10_part1/editbook/<int:id>, views.edit_book_lab10,
lab10_part1/deletebook/<int:id>, views.delete_book_lab10,

```

```

EXPLORER
  CS471...
  apps
    templates
      bookmodule
        booklist.html
        formatting.html
        index.html
        lab9_task1.html
        lab9_task2.html
        lab9_task3.html
        lab9_task4.html
        lab9_task5.html
        lab9_task6.html
        lab10_list.html U
        links.html
        list_books.html
        listing.html
        one_book.html
        search.html
        show.html

urls.py .../bookmodule M
lab10_list.html U
task7.html
urls.py libraryproject

1 <h2>Books</h2>
2
3 <a href="/books/lab10_part1/addbook">
4   Add Book
5 </a>
6
7 {% for book in books %}
8
9 <p>
10   {{ book.title }} - {{ book.author }}
11
12   <a href="/books/lab10_part1/editbook/{{ book.id }}">
13     Edit
14   </a>
15
16   |
17
18   <a href="/books/lab10_part1/deletebook/{{ book.id }}">
19     Delete
20   </a>
21 </p>
22
23 {% endfor %}

```



Books

[Add Book](#)

Thing, The - Jeniffer Lambirth [Edit](#) | [Delete](#)

K-20: The Fiend with Twenty Faces - Diana Toplis [Edit](#) | [Delete](#)

Secrets of the Tribe - Griff Winspur [Edit](#) | [Delete](#)

Unknown - Kakalina Szymczyk [Edit](#) | [Delete](#)

Coming Home - Rafaela Von Welldun [Edit](#) | [Delete](#)

There Was a Father (Chichi ariki) - Hayley Marieton [Edit](#) | [Delete](#)

Red Like the Sky (Rosso come il cielo) - Timoteo McDonand [Edit](#) | [Delete](#)

Teenage Mutant Ninja Turtles - Harlene Vassel [Edit](#) | [Delete](#)

Thing, The - Jeniffer Lambirth [Edit](#) | [Delete](#)

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Teenage Mutant Ninja Turtles - Harlene Vassel [Edit](#) | [Delete](#)

All Is Lost - Leonhard Cumes [Edit](#) | [Delete](#)

Apache Territory - Anders Coultard [Edit](#) | [Delete](#)

Police State - Fiann Guiso [Edit](#) | [Delete](#)

Godfather, The - Lynnett Leglise [Edit](#) | [Delete](#)

Miss Pettigrew Lives for a Day - Aurthur Castagnier [Edit](#) | [Delete](#)

Task 2: Create a URL '/books/lab9_part1/addbook' with any necessary HTML file and view function to insert the data of a new book.

```

175 def add_book_lab10(request):
176
177     if request.method == 'POST':
178
179         title = request.POST['title']
180         author = request.POST['author']
181
182         Book.objects.create(
183             title=title,
184             author=author
185         )
186
187     return render(request, 'bookmodule/lab10_addbook.html')

```

```

1 <h2>Add New Book</h2>
2
3 <form method="POST">
4
5     {% csrf_token %}
6
7     <label>Title:</label>
8     <input type="text" name="title">
9     <br><br>
10
11     <label>Author:</label>
12     <input type="text" name="author">
13     <br><br>
14
15     <input type="submit" value="Add Book">
16
17 </form>

```

127.0.0.1:8000/books/lab10_part1/addbook

Add New Book

Title:

Author:

Task 3: Create a URL `'/books/lab9_part1/editbook/<id>'` with any necessary HTML file and view function to update the data of the selected book.

```
def edit_book_lab10(request, id):  
    book = Book.objects.get(id=id)  
  
    if request.method == 'POST':  
        book.title = request.POST['title']  
        book.author = request.POST['author']  
  
        book.save()  
  
    return render(request,  
                  'bookmodule/lab10_editbook.html',  
                  {'book': book})
```

```
EXPLORER task3.html task2.html views.py M lab10_editbook.html U X  
apps > templates > bookmodule > lab10_editbook.html > form  
1 <h2>Edit Book</h2>  
2  
3 <form method="POST">  
4  
5     {% csrf_token %}  
6  
7     <label>Title:</label>  
8  
9     <input type="text"  
10         name="title"  
11         value="{{ book.title }}">  
12  
13     <br><br>  
14     <label>Author:</label>  
15  
16     <input type="text"  
17         name="author"  
18         value="{{ book.author }}">  
19  
20     <br><br>  
21     <input type="submit"  
22         value="Save Changes">  
23  
24 </form>
```

127.0.0.1:8000/books/lab10_part1/editbook/1

Edit Book

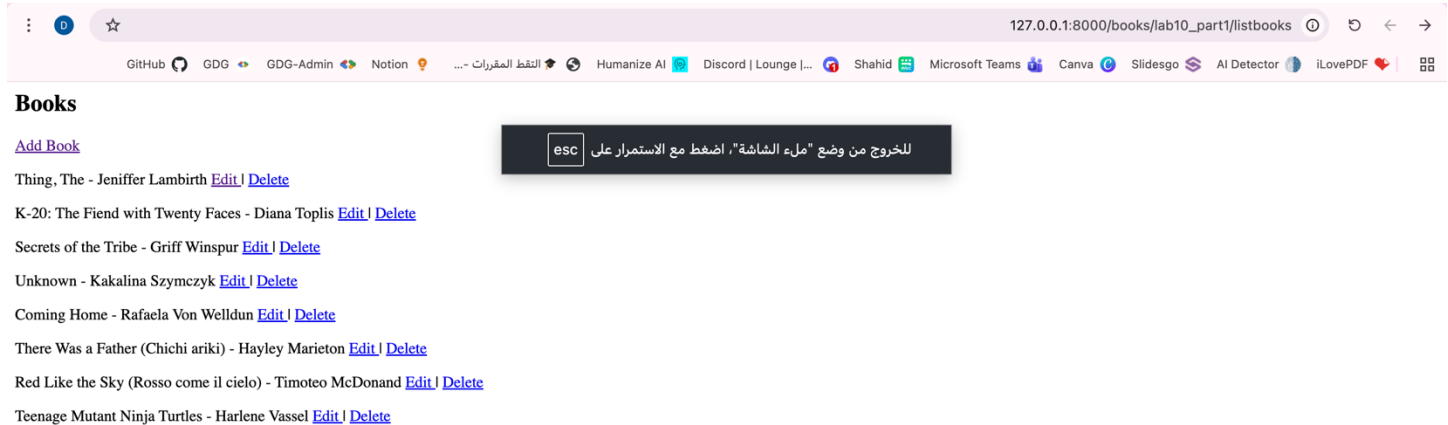
Title:

Author:

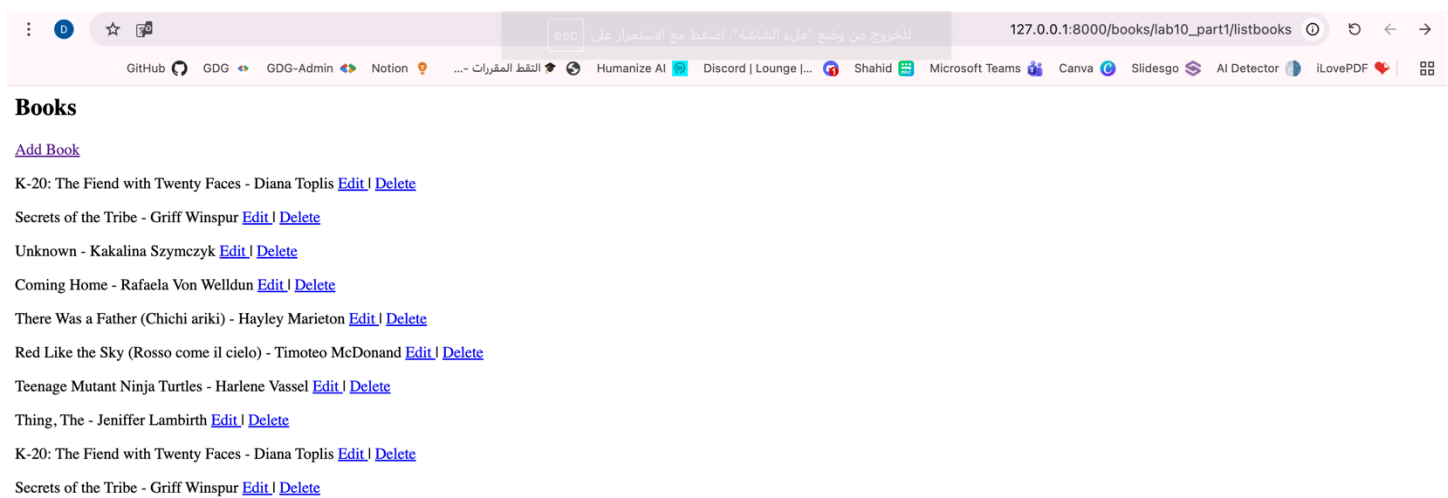
Task 4: Create a URL `'/books/lab9_part1/deletebook/<id>'` with a view function to delete the book with the specified id.

```
def delete_book_lab10(request, id):  
    book = Book.objects.get(id=id)  
  
    book.delete()  
  
    return redirect('/books/lab10_part1/listbooks')
```

Before delete:

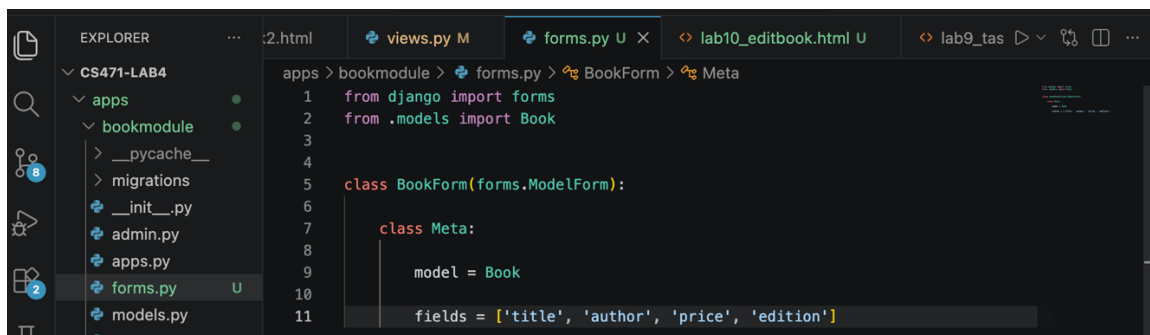


After delete(Thing, The):



Part 2: CRUD operations with validation (using Django forms)

Task 1: Reimplement all tasks in Part 1 using Django forms with URLs starting as '/books/lab9_part2/...'




```
booklist.html 210
formatting.html 211
index.html 212
lab9_task1.html 214
lab9_task2.html 215
lab9_task3.html 216
lab9_task4.html 217
lab9_task5.html 218
lab9_task6.html 219
lab10_addb... U 221
lab10_editb... U 222
lab10_list.ht... U 223
links.html 224
list_books.html 225
listing.html 226
one_book.html 227

def add_book_form(request):
    if request.method == 'POST':
        form = BookForm(request.POST)

        if form.is_valid():
            form.save()

            return redirect('/books/lab10_part1/listbooks')

        else:
            form = BookForm()

    return render(request,
                  'bookmodule/book_form.html',
                  {'form': form})
```

```
index.html 34
lab9_task1.html 35

path('lab10_part2/addbook', views.add_book_form),
]
```

```
EXPLORER
CS471...
apps
bookmodule
admin.py
apps.py
forms.py U
models.py
tests.py
urls.py M
views.py M

urls.py .../bookmodule M
book_form.html U
lab10_list.html U
lab10_addbc

apps > templates > bookmodule > book_form.html > form
1 <h2>Add Book Using Django Form</h2>
2
3 <form method="POST">
4
5     {% csrf_token %}
6
7     {{ form.as_p }}
8
9     <input type="submit" value="Save">
10
11 </form>
```



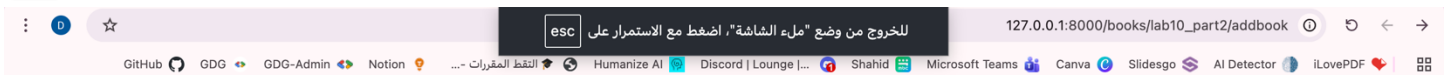
Add Book Using Django Form

Title:

Author:

Price:

Edition:



Add Book Using Django Form

Title:

Author:

Price:

Edition:



Books

[Add Book](#)

K-20: The Fiend with Twenty Faces - Diana Toplis [Edit](#) | [Delete](#)

Secrets of the Tribe - Griff Winspur [Edit](#) | [Delete](#)

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Police State - Fiann Guiso [Edit](#) | [Delete](#)

Godfather, The - Lynnett Leglise [Edit](#) | [Delete](#)

Miss Pettigrew Lives for a Day - Aurthur Castagnier [Edit](#) | [Delete](#)

Crimetime - Genia Devas [Edit](#) | [Delete](#)